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RESEARCH

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ATGAN: attention-based temporal GAN for EEG data augmentation in personal identification

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Abstract

With the development of brain–computer interfaces in recent years, deep learning models have been widely used in EEG-based personal identification. The training of complex models requires a large amount of data, but the acquisition of EEG signals is very time-consuming and energy-consuming. EEG data augmentation can reduce the overhead of EEG data acquisition and provide the data volume required by various complex models, which is conducive to applying EEG identification in practice. The generative adversarial network (GAN) has been applied to augment EEG data features. But the augmentation of temporal EEG signals is less studied. Therefore, this paper proposes a method of temporal EEG data augmentation based on an attention mechanism time series generative adversarial network (ATGAN), which can expand the original EEG dataset to facilitate the training of downstream deep learning classifiers. The experiment is conducted on the BCI Competition IV 2a dataset. The ATGAN model is trained with the temporal EEG signals after preprocessing, and the generated augmentation data by ATGAN is used to expand the original dataset. The accuracy of the EEG-based identification task on the augmented dataset is 7.78% higher than that on the original dataset, which proves the effectiveness of the proposed method.

Keywords: Temporal EEG, Data augmentation, Personal identification, ATGAN, Attention mechanism

1 Introduction

Due to its concealment, stress resistance, vividness, and revocation, the EEG signal is considered a safer biometric factor than traditional biometrics. In recent years, many researchers have paid attention to it and made great progress [1]. However, at present, most EEG datasets only involve a few hundred subjects at most. With the development of various identification algorithms, the algorithms are more complex and achieve high identification accuracy on these datasets, but this does not imply that these algorithms can maintain high accuracy on more subject datasets. Therefore, an important challenge for EEG authentication at present is to obtain enough data from more subjects, which puts forward a higher requirement for existing authentication algorithms, and training these algorithm models also requires a large amount of data.

The limited number of subjects in the datasets is because the acquisition of EEG signals requires professional brain–computer interface devices, which are not only expensive and cumbersome to wear but also need specific evoking paradigms during the signal acquisition process to stimulate the generation of related EEG signals. Generally, this requires the help of assistants to wear EEG caps or other acquisition devices to collect EEG signals and requires users to highly cooperate with the acquisition paradigm. Due to the low signal-to-noise ratio (SNR) of EEG signals, the signal acquisition process usually requires users to perform the same task several times to facilitate signal correction and denoising. This makes EEG data acquisition time-consuming and laborious, which poses a great challenge to the practical application of EEG authentication [2] and is more difficult for users to accept. Therefore, it is crucial to augment EEG data. For example, the BCI system uses the generation model to generate artificial data to expand the training set, thus improving the training speed of the BCI decoder and the generalization ability of the decoder [3].

Due to the collection equipment and paradigm, EEG signals are naturally temporal. Since the strong temporal variability and instability of EEG signals, generating the original temporal form of EEG signals is a challenge. However, the original temporal form of EEG signals is very important, and its augmentation is significant for subsequent processing of EEG signals. Due to the need for preprocessing such as denoising and artifact removal during the specific use of EEG signals, temporal EEG in this article refers to the EEG data after the original EEG signal has been preprocessed because various features of EEG signals are extracted and analyzed based on the original temporal form of EEG signals. Whether it is common AR features in the time domain, power spectral density features in the frequency domain, spatial domain, or nonlinear analysis, these all need to be analyzed and extracted from temporal EEG data. It can be said that temporal EEG data includes all useful features in the collected data.

In recent years, GAN [4] has developed rapidly in the field of image generation. It can effectively learn the distribution features of real images and generation image samples with the same distribution. Meanwhile, it can generate images that cannot be distinguished by human vision and use the augmented dataset to improve the performance of the classifier and make the classifier fit faster [5]. Some characteristics of EEG signals have been characterized as images by researchers and augmented by GAN. However, few studies have been conducted on the learning and augmentation of the time series data of EEG. This paper believes that it is significant to generate temporal EEG data for augmentation.

Temporal features are completely different from image spatial features, as they focus more on the relationship between the preceding and following moments. Jinsung Yoon et al. proposed a time series generative adversarial network to fully learn the time series characteristics of time series samples [6], and they verified the validity of the generated time series data on the sine series, stocks, home appliance energy, and lung cancer paths datasets. However, the generation model has not been applied to the generation of EEG data. Compared with these time series data, EEG signal has a high time resolution, low SNR, high non-stationary, and a certain degree of randomness. Generally, time series are unidimensional, and EEG signals not only have multiple channel dimensions, but also have correlation relationships.

This article draws inspiration from the idea of time series generative adversarial network and proposes ATGAN for the generation of temporal EEG signals. Firstly, attention mechanism is used to focus on the short-term temporal features and electrode channel relationships in the temporal EEG signals. Then, in addition to the traditional GAN generator and discriminator, an attention-based embedding is added to map the EEG features to a latent space, where the attention-based generator and discriminator are also trained. After all of the training are completed, the latent space features generated by the generator are input into the recovery and reconstructed into the temporal EEG. Time series generative adversarial network processes temporal signals into static and dynamic features, while EEG signals find it difficult to have stable static features. Therefore, ATGAN abandons static features and only focuses on the dynamic features of EEG signals. But unlike time series generative adversarial network mapping to low-dimensional space, our framework maps the representation latent space of EEG to high-dimensional space to more comprehensively represent the information of EEG based on the characteristics of EEG signals. Then, the generated data are expanded into the original dataset, thus meeting the data volume of identification classifier training and reducing the real data and time required. In this way, the practicality and user-friendliness of EEG-based identification in practical applications can be improved.

2 Related work

The traditional EEG data augmentation method is generally the same as the digital image augmentation method. The temporal data are geometrically transformed without changing the frequency, space, and power components [7]. For example, noise is added to the temporal EEG data sample, and the spatial electrode distribution of the EEG signal is rotated and deformed to generate the augmented EEG data, which is similar to the affine or rotational deformation of the image [8]. In this way, artificial EEG data samples can be also generated by combining and distorting the original experiment.

Window sliding is also a commonly used data augmentation method. It does not only change any data content, but only intercepts the data by sliding forward according to a certain window size in the time dimension. Fan et al. expanded the preprocessed EEG signal with a sliding window of 0.5 s, and they verified the augmentation effect on the PhysioNet dataset. Using this data augmentation method, it only takes 0.5 s of resting-state EEG after registration to realize rapid identification. When only 14 EEG channels and 0.5-s data are used, the average accuracy and average equal error rate of the system reach 99.32% and 0.18%, respectively. Compared with the existing work, this system has the advantages of short acquisition time, low computational complexity, rapid deployment, and using inexpensive EEG sensors in the market, which further promotes the implementation of the actual identification system based on EEG [9].

Piplani et al. exploited the improved loss function GAN to generate the power spectrum characteristics of EEG data and used it to augment the training dataset. With the augmented dataset, the accuracy of the classification model is improved from 90.8 to 95.0% [10]. Hartmann et al. proposed the first GAN framework called EEG-GAN for generating EEG signals, which improves the training of Wasserstein GANs to stabilize training. It can generate data close to real samples. However, it can only generate single-channel EEG data, and there are no specific application scenarios to prove it [11]. Abdelfattah et al. proposed recurrent GAN (RGAN) by replacing the full connectivity layer in GAN with RNN, which can capture the temporal characteristics in EEG signals [12]. Mirza et al. proposed CC-WGAN-GP and added a classifier (C) to WGAN-GP to improve GAN's ability to classify EEG data and make the generated samples more suitable for classification models [13]. Harada et al. employed the GAN based on the long short-term memory network (LSTM) to generate EEG time series data. The generator and the discriminator in the GAN adopt the multi-layer LSTM to capture the timing features in EEG signals. Besides, in [14], an EEG epilepsy detection dataset was used to train GAN, and the generated data were used to expand the original dataset, which can effectively improve classification accuracy.

Zhang et al. used DCGAN to generate spectrogram images of EEG signals, which were used to expand the spectrogram image dataset of the motor imagery (MI) classification task. The spectrogram images generated by FID quantitative analysis were more similar to the spectrogram images obtained by real EEG data processing than those generated by other methods. Further, as verified by the EEG MI classification task, the accuracy of the classifier on the dataset expanded by the generated image was significantly improved compared with that on the original dataset [15].

Zhang et al. proposed a framework called depth adversarial data augmentation (DADA), which is used to generate new data and enable the depth network classifier to train on small datasets. The BCI competition dataset contains three channels (C3, CZ, and C4) of 400 MI task experiments. The time–frequency characteristics of these EEG signals were extracted, and each EEG signal formed a $32 \times 32 \times 3$ image for training the CNN classifier. Compared with the traditional GAN, due to the redesigned loss function, the data augmentation of deep confrontation can generate more diversified data. By using the traditional CNN as a classifier and using the depth augmentation generation model to expand the EEG dataset, the accuracy can be improved from 74.8 to 79.3% [16].

Luo et al. proposed an EEG data augmentation framework based on conditional Wasserstein GAN (CWGAN). In CWGAN, the gradient penalty Wasserstein GAN algorithm is adopted to generate artificial EEG data in the form of differential entropy (DE) according to the noise distribution. Compared with the distorted geometric transformation, CWGAN can learn a deeper representation of real data distribution. The experimental results indicate that adding the training data generated by CWGAN can improve the accuracy of the EEG-based emotion recognition model [17]. Zhang et al. proposed a multiple generator cWGAN (MG-CWGAN) EEG data augmentation method based on cWGAN. They used the DE features selected by LDA on the SEED dataset to train MG-CWGAN, and the generated augmented data was used for classifier training. Compared with the cWGAN training, the proposed model converged faster and performed better on the augmented dataset using SVM as the classifier [18].

Zhang et al. proposed to use cDCGAN to generate time–frequency features of EEG signals in the form of images to expand the original time–frequency feature dataset of EEG. cDCGAN can be used to train the classifier of MI tasks and effectively improve classification accuracy [19]. Luo et al. proposed a conditional boundary equilibrium GAN (CBEGAN) method for data augmentation based on a genetic algorithm. The method was used to generate DE features of EEG signals. Its main advantage is that it overcomes the instability of the traditional GAN and has a fast convergence rate. The proposed method was evaluated on two multimodal emotion datasets. The experimental results indicate that the average accuracy of this method in three emotion classification tasks and five emotion classification tasks is improved by 4.6% and 8.9%, respectively [20]. The related works of using a depth generation model to augment EEG data are summarized in Table 1.

From the current research, EEG data augmentation is mostly in the format of the spectrogram, entropy, time–frequency feature, or power spectrum, while the research of using a depth generation model to augment temporal EEG signals and use them for EEG identification is very limited. This is because most of the current GAN models are developed in the field of digital images, but the most obvious feature of the original EEG signal is its high temporal resolution. To this end, this paper proposes a data generation method for temporal EEG signals to be used in EEG-based identification. By using an attention-based time series generative adversarial network (ATGAN), the features of temporal EEG signals are learned, and data with the same feature distribution as the real EEG data are generated to expand the dataset for identification.

Table 1 Related work of using a depth generation model to augment EEG data

Authors	Datasets	Classification task	Augmented format	Generation model	Accuracy (original)	Accuracy (augmented)
Tanya Piplani	Audio Video Evoked EEG Dataset	Personal authentication	Power spectrum	GAN	90.8%	95.0%
Harada	Epileptic Seizure Recognition Dataset	Epileptic Seizure Recognition	Raw EEG	LSTM-based GAN	NA	NA
Zhang	BCI competition IV dataset 1	MI classification task	spectrogram images	DCGAN	$74.5 \pm 4.0\%$	$83.2 \pm 3.5\%$
	BCI competition IV dataset 2b				$80.6 \pm 3.2\%$	93.2 ± 2.8
Zhang	BCI Competition IV 2b	MI classification task	Time–frequency images	DADA	74.8%	79.3%
Luo	SEED	Emotion recognition	DE	cWGAN	83.99%	86.96%
	DEAP			cWGAN	69.02%	78.17%
Zhang	SEED	Emotion recognition	DE	MG-CWGAN	NA	NA
Zhang	BCI competition II dataset III	MI classification task	Time–frequency images	cDCGAN	NA	NA
Hartmann	NA	Motor classification task	spectrogram images	WGAN	NA	NA
Luo	SEED	Emotion recognition	DE	CBEGAN	81.90%	87.56%
				cWGAN	81.90%	83.38%

3 Methods

ATGAN consists of four parts, namely an embedding function, recovery function, generator, and discriminator, all of which are implemented based on LSTM. An attention mechanism module has been added to the embedding function, which maps real EEG signals to high-dimensional latent space. The recovery function reconstructs latent space features to recover real EEG data, and an attention mechanism module is also added to the generator to generate latent space features. The discriminator determines whether it is real or generated. These four parts are jointly trained under the same latent space association. After training, the trained generator generates latent space features, and the generated EEG data are reconstructed by recovery. Figure 1 shows the overall framework of our ATGAN for temporal EEG data generation.

Common time series data include weather, rainfall, stock data, or data collected from monitoring devices such as sensors over a period of time. Time series data can be abstracted as invariant static features and time series features. In their study, static features refer to features such as gender, age, or indoor carbon dioxide concentration that remain unchanged over time and do not vary with time, and static features and time series features are extracted respectively through TGAN. However, EEG signals are non-stationary and random; the state of the brain is always different, and the corresponding EEG signals also differ from time to time.

Therefore, the EEG signals of the temporal data show almost no static characteristics in the time dimension. Although the relationship between EEG signal acquisition electrodes is determined by the acquisition equipment, once the data acquisition process starts, the position relationship between electrodes is fixed. However, since individuals have different brain states at all times, especially during different tasks, brain regions responsible for different tasks need to cooperate with each other. As a result, different brain regions have different levels of activity at different times and are highly sensitive to time. Determining the active brain regions corresponding to the electrodes at a certain time can help us to extract deeper information. Therefore, this paper will no longer model the static features in EEG signals, and in the proposed ATGAN, to further extract

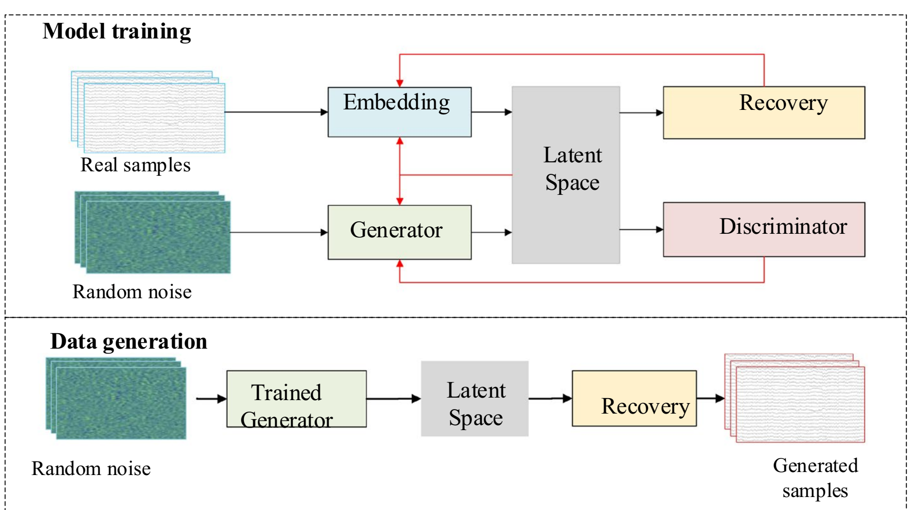

Fig. 1 Overall framework of our ATGAN for temporal EEG data generation

the influence relationship between temporal features and electrode channels in EEG signals, a feature extraction module based on the attention mechanism is added to the TGAN framework.

3.1 Embedding and recovery functions

The ATGAN extracts features from the original real data through the embedding and maps the extracted features to the latent space, while the recovery maps the features in the latent space back to the original real data. Meanwhile, the generator in the GAN generates the feature data of the latent space in adversarial learning. The temporal dynamic characteristics of samples are captured by ATGAN. Additionally, to capture more detailed timing information, the latent space of the embedding is mapped to high-dimensional latent space.

Let X be the original data space, x be the real sample in the data space, and H_X be the latent space mapped by X after embedding. The embedding e realizes $\prod_t \mathrm{X} \rightarrow \prod_t \mathrm{H}_X$, i.e., maps the real sample space and temporal features to high-dimensional latent space. The process can be formulated as

$$\mathrm{h}_{1:T} = e(X_{1:T}) \quad \mathrm{h}_t = e(h_{t-1}, x_t) \tag{1}$$

In this paper, e is implemented by a three-layer LSTM with an attention mechanism, where the attention mechanism layer is located after the second layer in the LSTM and before the third layer. The first two layers of LSTM perform shallow temporal feature extraction, followed by attention mechanism to further capture the relationship between temporal and channel dimensions. Then, the first layer of LSTM maps to 88 dimensions for representation in higher space. The attention mechanism is to allocate the information that needs to be considered in calculating the weight. The attention mechanism in this paper is divided into two parts. First, to pay more attention to temporal short-term characteristics, the attention mechanism of the time dimension is proposed, and the time dimension weight matrix W_1 is obtained through the time attention mechanism. This process can be expressed as

$$W_1 = R_S\left(mean\left(softmax(F_s(M_o))\right)\right) \tag{2}$$

where M_o represents the output of the second layer in the LSTM, $F_s(\cdot)$ represents the full-connection operation in the electrode channel dimension, $softmax(\cdot)$ represents the softmax function, and $mean(\cdot)$ represents the mean function. The corresponding relationship between the values of all electrodes in a sample at each time point is calculated. R_S stands for repeated filling in the electrode channel dimension, and finally, a 24×22 time series characteristic weight matrix W_1 is obtained. The degree of attention required at each time point is represented, and the operation of repeat is convenient for calculation. This process can be formulated as

$$W_2 = R_T\left(mean\left(softmax(F_t(M_o))\right)\right), \tag{3}$$

where $F_t(\cdot)$ indicates performing the full-connection operation in the time sequence dimension; R_T indicates that the chronological dimension is filled repeatedly. Finally, a

24×22 electrode channel dimension weight matrix W_2 is obtained. This process can be expressed as

$$M_i = M_o * (W_1 * W_2), \tag{4}$$

where M_i represents the output of the attention mechanism, i.e., the input of the third layer in the LSTM, which maps the output of the captured spatiotemporal features to a higher space to represent more information. The attention mechanism module in this paper is shown in Fig. 2.

For the recovery, a three-layer LSTM is trained to reconstruct the high-dimensional features generated by the embedding into the output data highly close to the real data. To reduce the complexity of the network, there is no attention mechanism module in the recovery, which will speed up the training process without affecting the performance of the recovery. The recovery implementation $\prod_t \mathrm{H}_X \rightarrow \prod_t \mathrm{X}$ reconstructs the coding features of high-dimensional space to the distribution of the original real data features. This process can be formulated as

$$\tilde{X}_{1:T} = \mathrm{r}(\mathrm{h}_{1:T}) \quad \tilde{x}_t = \mathrm{r}(\mathrm{h}_t) \tag{5}$$

To enable the embedding latent space feature distribution to accurately reconstruct the original data distribution through the recovery, a loss function $\mathcal{L}_\mathrm{R}$ is needed to minimize the distance between $\widetilde{x_t}$ and x_t as

$$\mathcal{L}_\mathrm{R} = \mathbb{E}_{\mathbf{X}_{1:T} \sim p}\left[\sum_t \parallel x_t - \tilde{x}_t \parallel_2\right]. \tag{6}$$

3.2 Generator and discriminator

Due to the limitations in learning temporal characteristics of the traditional GAN structure, the generator here will generate latent space characteristics corresponding to the embedding, thus better learning the temporal characteristics of data and channel dimension information. Let Z be the random vector space input by the generator, z be the sample in the random vector space, and generator g realizes $\prod_t \mathrm{Z} \rightarrow \prod_t \mathrm{H}_X$. After the

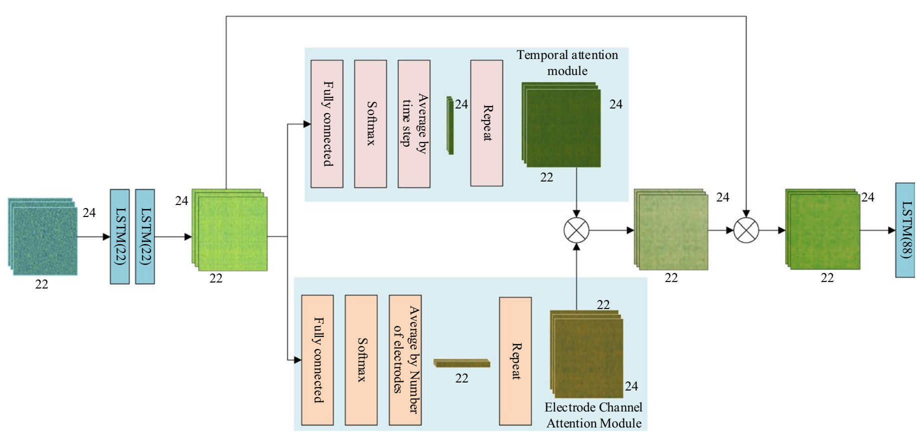

Fig. 2 Attention mechanism module in embedding and generator

generator, the vector space is mapped to the latent space with the same data distribution as the real sample H_X. This process can be expressed as

$$\tilde{\mathrm{h}}_{1:T} = g(Z_{1:T}) \quad \tilde{\mathrm{h}}_t = g(h_{t-1}, z_t). \tag{7}$$

To make the latent space feature generated by the generator has a consistent internal structure with the latent space feature, the structure of the generator is identical to that of the embedding. It is implemented by a three-layer attention mechanism LSTM, where the attention mechanism layer is located after the second layer of the LSTM and before the third layer. The attention mechanism module is consistent with that in the embedding.

The discriminator distinguishes latent space features corresponding to real samples or generated samples, and discriminator d realizes $\prod_t \mathrm{H}_X \to \prod_t [0,1]$, to complete the judgment of the real samples or generated samples corresponding to the latent space features. The discriminator is implemented by connecting a three-layer LSTM with a classifier d, and $\tilde{\mathrm{y}}_t$ represents the discrimination result of the discriminator. This process can be formulated as

$$\tilde{\mathrm{y}}_t = d\left(\underset{t}{\overleftarrow{u}}, \vec{u}_t\right), \tag{8}$$

where $\underset{t}{\overleftarrow{u}} = \overleftarrow{c}\left(\tilde{\mathrm{h}}_t, \underset{t+1}{\overleftarrow{u}}\right)$ and $\vec{u}_t = \vec{c}\left(\tilde{\mathrm{h}}_t, \vec{u}_{t-1}\right)$ represents the backward latent layer status and forward latent layer status of the last layer in the LSTM, respectively; $\overleftarrow{c}, \vec{c}$ represents the LSTM function of the entire three layers in the discriminator. The generator and discriminator are trained by an unsupervised confrontation loss. This process can be expressed as

$$\mathcal{L}_{\mathrm{U}} = \mathbb{E}_{\mathbf{x}_{1:T}\sim p}\left[\sum_t \log y_t\right] + \mathbb{E}_{\mathbf{x}_{1:T}\sim \hat{p}}\left[\sum_t \log\left(1-\hat{y}_t\right)\right]. \tag{9}$$

3.3 Joint training

Since the space mapped by the embedding and generator is the same high-dimensional latent space, the data space generated by the generator is consistent with the latent space mapped by the embedding. By minimizing the difference between the latent space features of the real sample mapping and the latent space features of the generated sample mapping, the generator and the embedding can achieve the same mapping distribution $p(H_X)$ and $\hat{p}(H_X)$ in the latent space during the joint training process. The overall structure of ATGAN is presented in Fig. 3.

Meanwhile, maximum likelihood estimation is exploited to optimize the supervision loss as

$$\mathcal{L}_{\mathrm{S}} = \mathbb{E}_{\mathbf{x}_{1:T}\sim p}\left[\sum_t \| \, h_t - g(h_{t-1}, z_t) \, \|_2\right], \tag{10}$$

where $\mathcal{L}_{\mathrm{S}}$ promotes the generator to further generate data sequences close to the real through the latent space feature supervision training corresponding to the real data. In

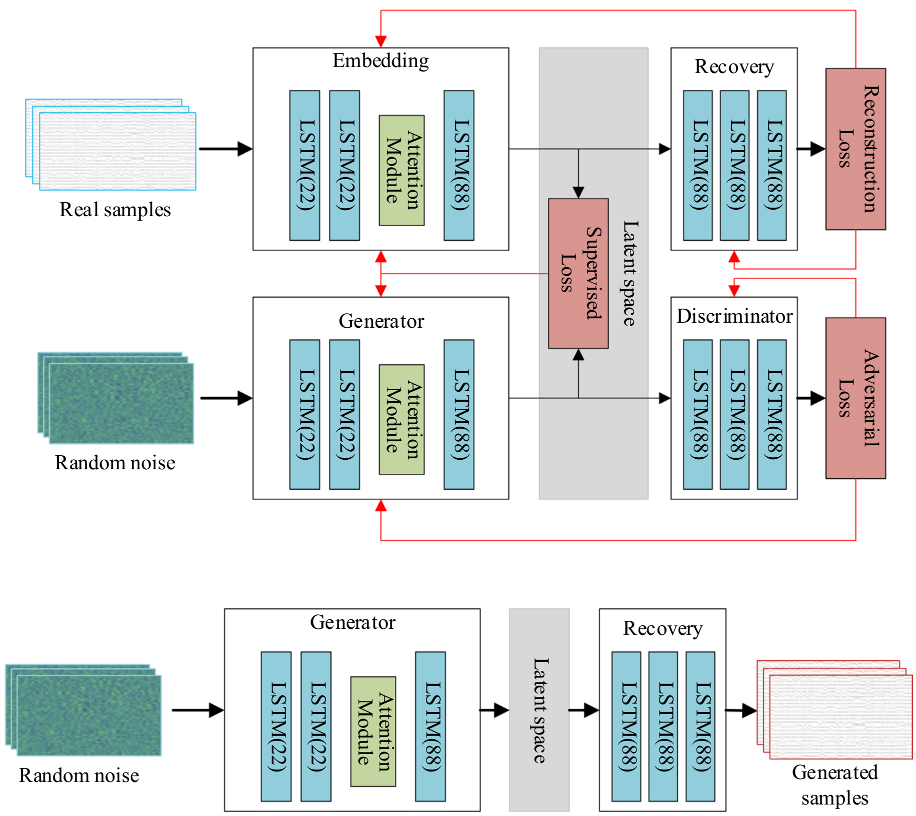

Fig. 3 Overall structure of ATGAN

fact, $\mathcal{L}_S$ connects the embedding with the generator in a real sense, and it continuously optimizes two mapping functions through a supervision loss to enable them to map the input data to the same space. Since the key to joint training is the latent space, the mapping of embedding is crucial as it directly determines the distribution of latent space features. Therefore, the attention mechanism is placed in embedding, and the generator also adds an attention mechanism module to generate features with the same distribution. The entire attention mechanism module only exists in embedding and generator.

4 Experimental results and discussion

4.1 Dataset: BCI competition IV 2a

This dataset contains four types of motor imagery data collected and recorded on nine healthy subjects. The motor imagery task includes four different guided spontaneous imagery tasks, namely, left-hand motor imagery, right-hand motor imagery, feet motor imagery, and tongue motor imagery. During data collection, the subjects sat on a comfortable armchair with a computer screen in front of them. At the beginning of the test, a fixed cross appeared on the computer screen with a short voice prompt. Two seconds later, an arrow pointing to the left, right, up, and down (corresponding to four movements respectively) appeared on the screen. The subjects completed the imagery tasks according to the arrows on the screen, with no feedback information. This dataset gives the continuous signals of 25 EEG channels (22 EEG signal channels and 3 EOG channels), and the acquisition frequency is 250 Hz, thus providing complete tag information.

Since this paper focuses on EEG signals, so these three EOG channels are discarded. The channel name and the corresponding position of the channel are shown in Table 2:

4.2 Data preprocessing

Since our task is personal identification, different ATGAN models are trained for each of the nine subjects in the dataset, generate their own data, and then augment the generated data into the original dataset. Although the dataset contains different tasks, including motor image. This paper does not consider the impact of motor imagery task and resting state on personal identification. The first 10,000 sampling points of each individual are selected to process as the training and testing dataset for ATGAN. The sampling frequency is 250 Hz, which means there are 250 sampling points per second, so the first 40 s of data collected for each subject are selected. Therefore, the data size selected for each subject is $(22 \times 10{,}000)$, which means channels = 22, time sequence = 10,000. According to the dataset description, the 40-s data include motor imagery tasks and resting states.

Since the EEG signal has a very low SNR, it is easy to mix with other noises, so it generally needs to be band-pass filtered to remove power frequency interference first. Then, the EEG signal is analyzed by independent component analysis (ICA), and ocular artifacts are deleted. In medical images, artifacts can be defined as any feature that distorts an object in the field of view (FOV) of an image. The electrode position of EEG signal acquisition is close to the eyes, and the myoelectric signals generated by blinking and eye activities are stronger than EEG signals. Removing these signals is important for obtaining clean EEG signals. In this paper, a notch filter is first used to remove power frequency interference at 50 Hz, and then a high-low pass filter of 0.1–45 Hz is employed to obtain the target frequency band. Next, ICA is conducted to draw the time series signal diagram of each component and remove the ocular electrical artifact component. The original EEG waveform and the denoised EEG waveform are illustrated in Fig. 4.

Because the above preprocessing, noise removal, and artifact removal operations need to be visualized, EEG data are processed in the form of a waveform. It can be seen that after preprocessing, obvious ocular artifacts are removed, the EEG data becomes smoother, and the noise is significantly reduced. However, the data are still stored in a discrete form according to the sampling points, and each point corresponds to the voltage value of each electrode. To obtain more samples, the sliding window method is adopted to generate the input samples needed for training the generation model. After the samples are obtained, the data samples are subject to min–max scaling to facilitate the training of the GAN model, which can make the influence

Table 2 Number of electrodes, electrode positions, and the number of people and sampling frequencies in the dataset

Dataset	Number of electrodes	Electrode position	Number of subjects	Sampling frequency
BCI Competition IV 2a	22	'Fz','FC3','FC1','FCz','FC2','FC4','C5','C3','C1','Cz', 'C2','C4','C6','CP3','CP1', CPz','CP2','CP4','P1','Pz', P2','POz'	9	250 Hz

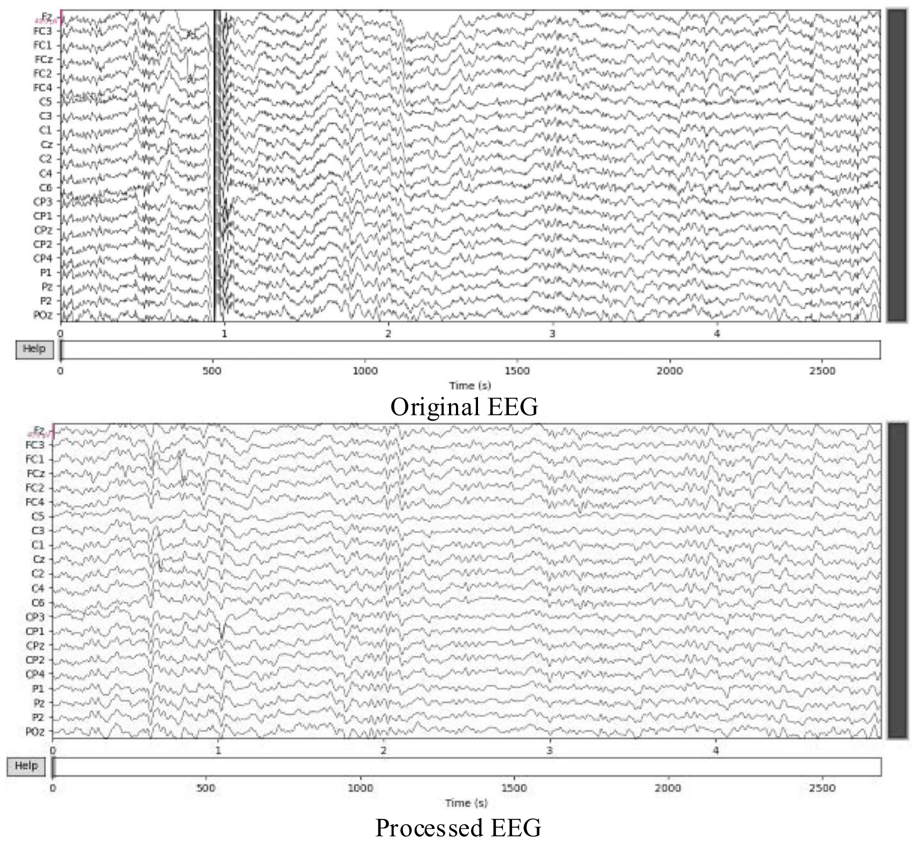

Fig. 4 Original EEG waveform and the denoised EEG waveform

weights of each feature dimension on the objective function consistent and improve the convergence speed of the model iterative solution.

Given a common dataset containing N sample points, denote the window length as L, the data channel dimension of EEG signals as D, and the sliding step size as S. Since the data channels collected in the dataset are fixed, D is constant, and the window size determines the size of a single sample. The sliding step S determines the number of samples that can be obtained under the same amount of data and the similarity between samples. The smaller the value of S is, the more samples are obtained through the sliding window, and the higher the similarity between the samples. Fan et al. proved through experiments that when the classifier is trained with samples cut from smaller sliding steps, the classification accuracy is higher than that of long sliding steps [10]. For a visual representation, oscillograms are used for the schematic diagram of sample interception using window sliding, as shown in Fig. 5.

This paper selects the first 10,000 sampling points for each subject, that is, $N = 10{,}000$, with a step size S of 1 and a window size $L = 24$. Therefore, each sample size is (22,24), representing data from 24 sampling points across 22 channels. The data size for each subject is (9976,22,24), which means samples number = 9976, channels = 22, time sequence = 24. Since this is an offline dataset, these 9976 pieces of data are randomly divided into training and test data according to a 4:1 ratio. The same

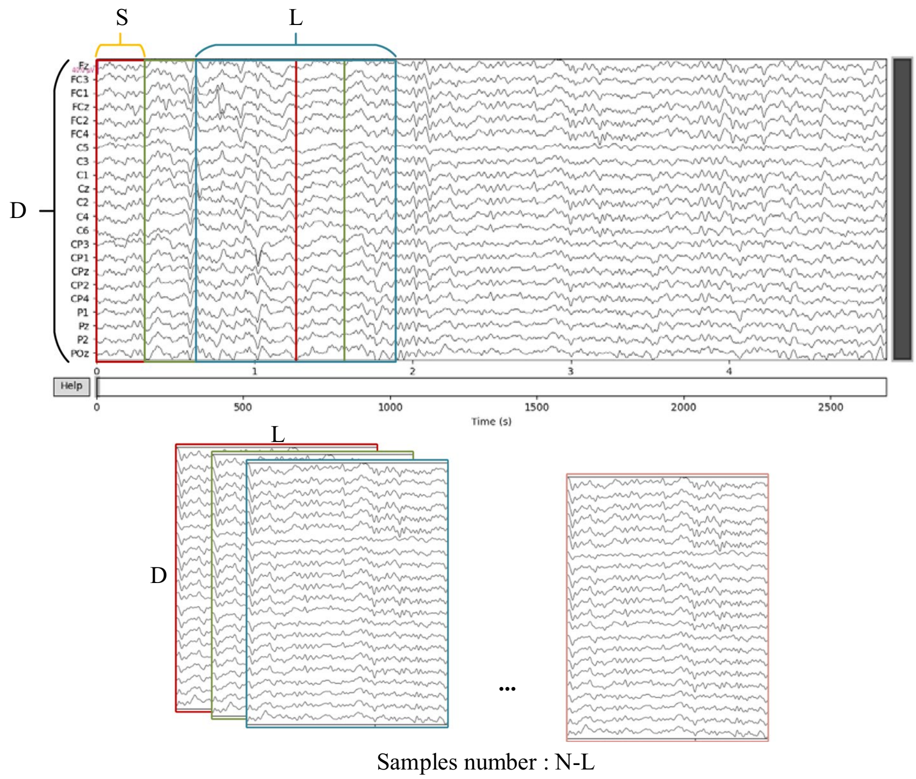

Fig. 5 Intercepting training samples by sliding the window

operation is performed on the data of each subject, and then the training data of the nine subjects are combined into a training set, and all test data are combined into a test set. The training set will be used to train the personal identification classifier and ATGAN. Since ATGAN is unsupervised learning, it will be trained separately for each of the nine subjects and generate data to augment the training sample set of each subject.

4.3 Experiments

This paper evaluates the proposed EEG generation method from three aspects: qualitative analysis, quantitative analysis, and the actual performance of downstream identification tasks. Qualitative analysis is to intuitively show the similarity between the generated data and the real data, while quantitative analysis is to accurately calculate the gap between the generated data and the real data. The downstream identification task is to determine whether the data generated by the proposed generation method is effective. Effective data augmentation can improve the performance of the classifier, especially the performance of the deep learning model. This is because the training process is data-driven, and the data size directly affects the training results.

After adopting the above data preprocessing method, the samples are intercepted from the original data by window sliding to form the training dataset of ATGAN and the original dataset of EEG identification.

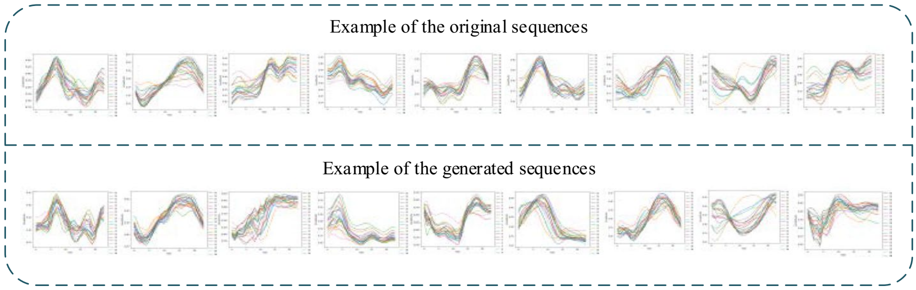

Fig. 6 Example of the original and generated sequences

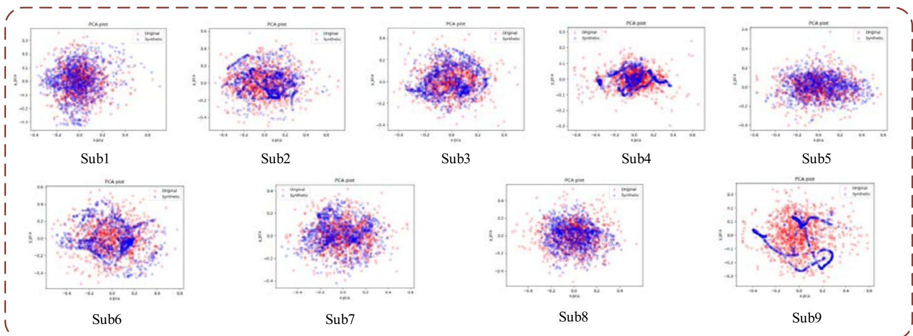

Fig. 7 PCA visualization of the data generated by ATGAN and the real data

(1) Qualitative analysis: visualization

Firstly, this paper presents sample examples of the generated data and real data using the proposed method in Fig. 6. The first row of image in Fig. 6 shows a real sample selected from nine subjects, while the second row of image shows the most similar generated sample calculated through cosine similarity among the corresponding subjects. The horizontal axis of the image is the time sampling point, which is 24, and the vertical axis is the amplitude. In order to highlight the amplitude changes, the data of 22 electrodes are displayed in the same coordinate. It is worth noting that these samples have been normalized, but they are all directly used as input samples for personal identification. It can be seen that the generated data are very close to the real data in terms of amplitude changes intuitively indicating that the generated data effectively learn temporal features from real data.

PCA and t-SNE are applied to the original dataset and the generated dataset for data visualization. This can intuitively and qualitatively show how close the distribution of the generated samples is to the distribution of the original samples in the two-dimensional space. The closer the distribution is, the more similar the generated data is to the original data distribution, and the more authentic the generated data is. But in fact, what works is the degree of overlap between the two distributions. The more the distribution of blue and red dots overlaps, the closer the generated data is to the real data distribution. The PCA and t-SNE visualization results of the data generated ATGAN models and real data are illustrated in Figs. 7 and 8, respectively.

In the visualization image, the red dot represents the distribution of the original data, and the blue dot represents the distribution of the generated data. It can be

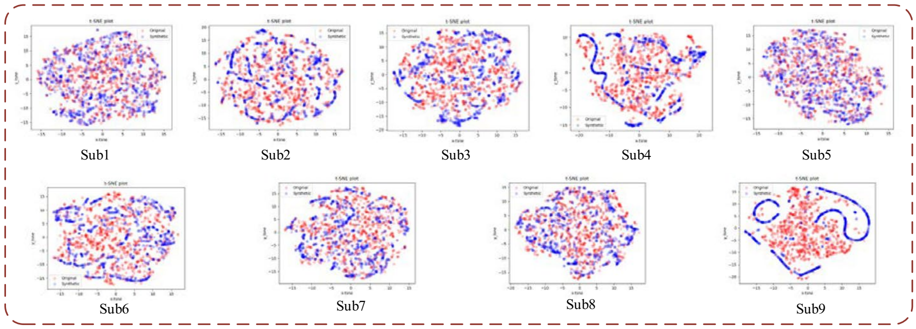

Fig. 8 T-SNE visualization of the data generated by ATGAN and the real data

seen from the PCA visualization result, the distribution of the EEG data generated by ATGAN is obviously close to the real data. The t-SNE visualization result indicates that the data generated by ATGAN can well fit the distribution of the real EEG data.

In the visualization image, it can be seen that the distribution of some generated data has undergone clustering or obvious trajectories, which to some extent indicates that ATGAN focuses on certain specific information in the raw data during training due to attention mechanisms. To verify whether this information is valid, further experiments are needed.

Although PCA and t-SNE visualization can intuitively reflect the sample distribution of the generated data and the real data, this only quantitatively analyzes the gap between the generated data samples and the real samples. This is also necessary for quantitative analysis and downstream task analysis. Especially for downstream classification tasks, personal identification experiments can be conducted after expanding to the original dataset to determine whether the generated data is effective and whether the data contains feature information, which will be directly reflected by the impact on the classification results.

(2) Quantitative analysis: discriminative score and predictive score

Quantitative analysis is to accurately measure the gap between generated data and real data. The paper compares ATGAN with GAN [10], DCGAN [15] and LSTM-GAN, which the first two are widely used methods in the field of image generation and EEG generation. In addition, LSTM-GAN [14] is the most closely related methods to ATGAN. Due to the methods involved in this paper are all unsupervised learning, while the DADA method is supervised learning, this paper will not compare with it.

Discriminative score: To accurately measure the gap between the generated data and the real data, a two-layer LSTM time series classification model is trained to classify the original dataset and the samples of the generated dataset. First, each original sequence is labeled real, while each generated sequence is labeled not real. Then, an LSTM classifier is trained to distinguish the two classes as a standard binary classification task. If the generated data is very real, the classifier cannot distinguish which is true and which is generated, so it can only randomly guess one. The accuracy should be 0.5. Therefore, this paper takes 0.5 as the ideal accuracy, and the discrimination score is calculated as

Table 3 Discriminative scores based on ATGAN and other methods (lower the better)

	1	2	3	4	5	6	7	8	9
GAN	0.4997	0.4499	0.4999	0.4996	0.4497	0.497	0.4979	0.5	0.4997
DCGAN	0.5	0.4964	0.5	0.4944	0.4736	0.4821	0.4939	0.5	0.5
LSTM-GAN	0.5	0.5	0.5	0.5	0.4944	0.4817	0.4787	0.4885	0.478
ATGAN	**0.1045**	**0.1863**	**0.1818**	**0.0364**	**0.1476**	**0.2204**	**0.1063**	**0.2585**	**0.2422**

Bold values indicate the best results under each evaluation metric

Table 4 Predictive scores based on ATGAN and other methods (lower the better)

	1	2	3	4	5	6	7	8	9
GAN	0.0504	0.0475	0.0641	0.0271	0.0471	0.0563	0.0612	0.0973	0.066
DCGAN	0.3817	0.0553	0.6219	0.0268	0.1064	0.0563	0.1313	0.197	0.5264
LSTM-GAN	0.3241	0.359	0.1773	0.1879	0.0221	0.0258	0.0334	0.0478	0.0358
ATGAN	**0.0133**	**0.0096**	**0.0141**	**0.0075**	**0.0103**	**0.0135**	**0.0159**	**0.0159**	**0.0240**

Bold values indicate the best results under each evaluation metric

$$\text{discrimination score} = \left|\text{accuracy} - 0.5\right|. \tag{11}$$

The lower the discrimination score, the closer the generated data is to the real data. According to the quantitative analysis, the discriminative scores of ATGAN are smallest, indicating that the gap between the generated data samples based on ATGAN and the real samples are smallest, and the generated data is closest to the real data. The discriminative scores of the data generated and the real data of 9 subjects based on ATGAN and other methods in the dataset are shown in Table 3.

Predictive score: In addition to measuring the similarity between the generated data samples and the real data samples, the ability of the generated data to inherit the characteristics of the original data is also measured. Because the data form is a time series, this experiment evaluated the performance of ATGAN in capturing the conditional distribution over time. Therefore, a two-layer LSTM time series prediction model is trained by using the generated dataset samples to predict the next step time feature vector of each input series. Then, the trained model is evaluated on the original dataset.

Mean absolute error (MAE) is used to measure the average absolute error of two sets of data, which can intuitively reflect the difference between the two sets of data. Its range of values is $[0, +\infty)$. The more similar the two sets of data are, the smaller the value of MAE and tends toward 0. The larger the difference, the larger the value of MAE. The MAE is calculated as

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n}\left|y_i - \hat{y}_i\right|. \tag{12}$$

The predictive scores of the data generated and the real data of nine subjects based on ATGAN and other methods in the dataset are shown in Table 4.

The predictive scores of ATGAN are also smallest, indicating that the generated data of ATGAN is consistent with the real data from the perspective of data distribution. The low predictive scores also indicate that the temporal features of the generated data have been fully learned. The discriminative score and the predictive score further indicate that ATGAN performs better in the generation of EEG temporal data.

(3) Downstream tasks: personal identification

The purpose of this paper is to expand the EEG identification dataset. To verify the validity of the generated data, the same EEG identification classification network is trained on the augmented dataset and the original dataset. The use of the trained model to generate data to expand the original dataset is shown in Fig. 9.

Generally, for the same dataset, the more complex the model is, the better the classification performance will be. That is, when training complex classification models on the datasets before and after expansion, the classification accuracy may not differ much. The classification algorithm used for EEG identification has been very mature and complex, and the accuracy is also very high [21]. To demonstrate the impact of the augmented dataset with generated data on the performance of the classifier, a simple two-layer LSTM network is taken as the EEG identification classifier. Note that unlike the classifier for the above discriminative score, the classifier here classifies each subject in the dataset to achieve personal identification. The above classifier is to distinguish the gap between generated data and the real data. The classifier trained with the augmented dataset and the original dataset is tested on the same test set. Then, the trained generation model is used to generate data of the same size as the original training set to expand the original dataset.

In the classification experiment, the training of the classifier adopts the same settings and parameters except for different datasets. Compared with the original dataset and the expanded dataset based on GAN, DCGAN, and LSTM-GAN methods, ATGAN can effectively improve the accuracy of the classifier. The identification

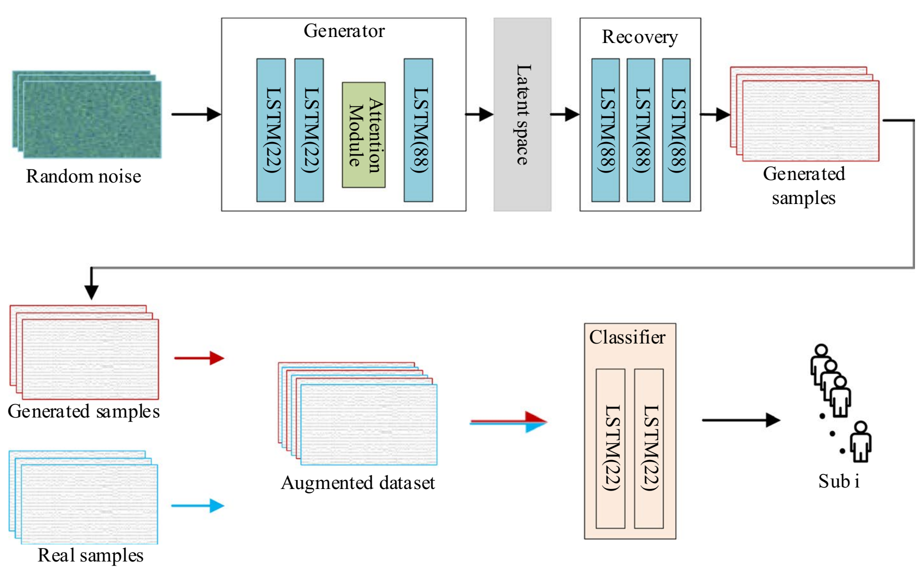

Fig. 9 Using the trained model to generate data to expand the original dataset

Table 5 Accuracy of the classifier on each subject in real dataset and augmented dataset

	1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**Total**
Original	83.61	90.61	85.21	97.78	99.70	97.84	98.88	90.27	78.59	91.49
GAN	83.10	91.10	82.52	98.39	**100.00**	95.88	99.49	95.81	84.01	92.26
DCGAN	91.35	92.91	99.04	98.69	**100.00**	93.32	97.82	93.81	89.54	95.14
LSTM-GAN	92.67	95.99	99.59	99.34	99.65	91.07	93.96	93.41	88.35	94.87
ATGAN	**97.10**	**99.41**	**99.75**	**99.65**	**100.00**	**99.85**	**99.95**	**99.00**	**98.71**	**99.27**

Bold values indicate the best results under each evaluation metric

accuracy of these three baseline models is unstable on each subject, with some having high accuracy while others have low accuracy, and even lower accuracy compared to original dataset. This is because of the low quality of samples generated on the subject, which leads to the identification model recognizing it as samples of other subjects, and the total number of samples increases, resulting in a decrease in accuracy. It shows that the generated data by ATGAN is more effective as a supplement to the original dataset. The accuracies of the personal identification on each subject on original dataset and GAN, DCGAN, LSTM-GAN, ATGAN-augmented datasets are shown in Table 5.

In the experiment part, this paper presents sample examples of the generated data and real data using ATGAN. PCA and t-SNE are used to visualize the generated data and the original data, thus intuitively and qualitatively showing the identification of the generated data. Then, the discriminative score and predictive score are used to quantitatively evaluate the closeness between the generated data and the real data. It can be seen that the performance of the data generated by ATGAN is pretty similar to real EEG data. Finally, the specific downstream classification task verification is conducted to verify the performance impact of the classifier training using the expanded dataset of generated data. The same classifier is trained on the ATGAN-augmented dataset and the original dataset, and ATGAN achieved the high classification accuracy, indicating that the data generated by ATGAN is pretty real and effective.

5 Conclusion

To solve the problem of difficult data acquisition in EEG identification, this paper proposes an EEG data generation model (ATGAN). Aiming at the non-stationary and random characteristics of EEG data, the data samples are processed into short-term sequence samples so that ATGAN pays more attention to the short-term time series characteristics. Meanwhile, the electrode dimension attention is added to make it more suitable for the characteristics of EEG data. ATGAN is trained on the samples in the BCI Competition IV 2a dataset, and the generated data are evaluated through qualitative analysis, quantitative analysis, and EEG identification tasks to verify that the data generated by ATGAN are pretty similar to real EEG data and effective. The proposed method can greatly generate a large amount of data to expand the dataset required for training classification models under limited data conditions. The method can reduce the amount of data required for EEG identification, reduce the collection time, and improve user-friendliness. In future work, we will further explore the effects of ATGAN on the generation of various special evoked EEG signals.

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Author contributions
Conceptualization was done by SZ, LS, XM, MZ and YH; methodology was done by SZ, LS and XM; software was done by SZ, LS, XM, MZ and YH; validation was done by SZ, LS, XM, MZ and YH; formal analysis was done by SZ, LS and XM; investigation was done by SZ, LS and XM; resources were done by SZ, LS, XM, MZ and YH; data curation was done by SZ; writing—original draft preparation was done by SZ, LS, XM, MZ and YH; writing—review and editing was done by SZ, LS and XM; visualization was done by SZ, LS, XM, MZ and YH; supervision was done by SZ, LS and XM; project administration was done by SZ and MZ; funding acquisition was done by SZ, LS and XM. All authors have read and agreed to the published version of the manuscript.

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Availability of data and materials
The code used in this study is available upon request from the corresponding author. The datasets that support the findings of this study are openly available at [https://www.bbci.de/competition/iv/.].

Declarations

Competing interests
The authors declare no competing interests.

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